

Rafik El Khoury

AI in the Built Environment

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Summary

Civil engineer, urban planner, and AI researcher exploring the role of AI in the Architecture, Engineering, and Construction industry. Currently at IAAC building tools for the built environment — automated compliance checking, project-manager communication pipelines, and augmented regenerative design. Looking for a team where I can take ideas from concept to product. Authorized to work in the **UK** (Graduate Visa), **EU** (Greek citizenship), and **UAE** (Golden Visa).

Projects

NatureGooddest / COOLSTOCK — Regenerative Design Platform 2026

Pattern-language platform that turns a site polygon into a defensible urban-design brief — every recommendation traceable to a theory citation, a verified data source, and a field-validation record. v1 (**COOLSTOCK**) pitching Ajuntament Barcelona's Refugis Climàtics programme, June 2026.

- v1 on Plaça dels Àngels, Barcelona (3,800 m², Tmrt 58 °C) — 27 patterns, NSGA-II Pareto-front shelter configs, 5-regulation compliance card (EU Taxonomy, CSRD, TNFD, BNG).
- Authored the **three-citation rule** (theory DOI + data provenance + field record) and the deterministic generation path; LLM is editor/search only, never in the design loop.
- Stack — Python/Flask, Neo4j + OWL/Turtle KG, NSGA-II (pymoo), Infrared.city CFD, Open Data BCN, pgvector. Team of 4 at IAAC MaAI01, Dir. Emanuele Naboni.

GigAI — Material Change Coordinator for Construction PMs 2026

Event-driven workflow automation that captures construction material changes from meetings and PM tools, then generates coordinated, PM-approvable actions with confidence scoring.

- Pipeline — Fireflies transcripts → event normalization (Claude Haiku) → context enrichment (pgvector RAG over floor plans, suppliers, history) → proposal generation (Claude Sonnet) → ranked actions for PM approval.
- Integrated Autodesk Construction Cloud, Google Calendar, Gmail; React+Vite dashboard with SSE for real-time decisions. 4-factor confidence scoring lands demo proposals at ~85%.
- Stack — Python/FastAPI, React+Vite, PostgreSQL+pgvector, Docker, GCP Pub/Sub, Anthropic Claude (Haiku + Sonnet), Voyage AI embeddings.

ARCHAI — Automated Compliance Checking for Urban Regulations 2025 – 2026

Rhino-integrated tool that checks building geometry against urban planning regulations using vector-store retrieval and LLM-generated explanations.

- Pipeline — Rhino building geometry → Chroma vector-store rule retrieval (zone, metric, operator, threshold, source) → per-rule pass/fail table with article-cited LLM explanations.
- Tested against city regulation corpora (Mumbai, Barcelona) with a consistent rule-schema so the engine is regulation-agnostic.
- Stack — Python, ChromaDB (RAG), OpenAI GPT-4o-mini, Pandas, n8n; Rhino interop via CSV manifests.

Experience

Graduate Urban Planner, Rafik El-Khoury & Partners — Consulting Engineers – Abu Dhabi, UAE Jan 2025 – Sept 2025

- Evaluated international masterplan proposals for a 19 km² island development; built a scoring framework weighting design merit, consultant expertise, and long-term vision alignment.
- Contributed to SHA villa complex redesign in Al-Jurf — piling works and peer review, CAD refinement, and real-estate sales brochure design.
- Programmed three community parks in Madinat Al-Riyadh through site analysis and public space planning.
- Authored UNICEF-guided school rehabilitation proposals in southern Lebanon.
- Briefed the president of Aramoun municipality on the town's urban character, planning issues, and proposed interventions.

Site Engineer, IMKAN Properties – United Arab Emirates Jan 2023 – Mar 2023

- Conducted villa inspections across multiple construction stages and produced weekly progress reports for all villas.
- Worked across construction sequences and methods, including tiling techniques and waterproofing applications.

Engineering Intern, Rafik El-Khoury & Partners — Consulting Engineers – Beirut, Lebanon Aug 2020 – Sept 2020

- Reviewed the Solid Waste Master Plan for Greater Beirut — assessed waste generation, environmental impact, and recycling strategies for sustainability and regulatory compliance.
- Supervised on-site activities at the Burj Hammoud Landfill — leachate collection oversight, waste-load recording, and worker safety procedure enforcement.

Education

Institute for Advanced Architecture of Catalonia (IAAC), MS in AI in Architecture and the Built Environment – Barcelona, Spain Oct 2025 – present

- Expected graduation July 2026.

University of Dundee, MSc in Spatial Planning with Sustainable Urban Design – Dundee, Scotland Sept 2023 – Sept 2024

- Dissertation on Natural Flood Management (biodiversity and flood resilience in England); team-based research-by-design masterplan for Dundee's urban-rural fringe.

Quinta do Vale, Cert in Permaculture Design – Portugal July 2022 – Aug 2022

Loughborough University, BEng in Civil Engineering – Loughborough, UK Oct 2018 – June 2022

Skills

AI/ML & Programming: Python, JavaScript/TypeScript, SQL; LLM APIs (OpenAI, Anthropic, Gemini), RAG (ChromaDB, pgvector), Voyage AI embeddings, Neo4j + OWL/Turtle knowledge graphs, NSGA-II multi-objective optimization (pymoo)

Backend / Frontend: FastAPI, Flask, React+Vite, PostgreSQL, Docker, Google Cloud Pub/Sub, n8n, Server-Sent Events, Git

Design & Engineering: Rhino, Grasshopper, AutoCAD, BIM, Ladybug environmental analysis, CFD integration (Infrared.city), IFC parsing

Planning & GIS: GIS, spatial analysis, Open Data Barcelona (CKAN), EPSG coordinate systems, statutory planning, report writing

Languages: English (fluent), French (fluent), Arabic (native), Spanish (conversational)